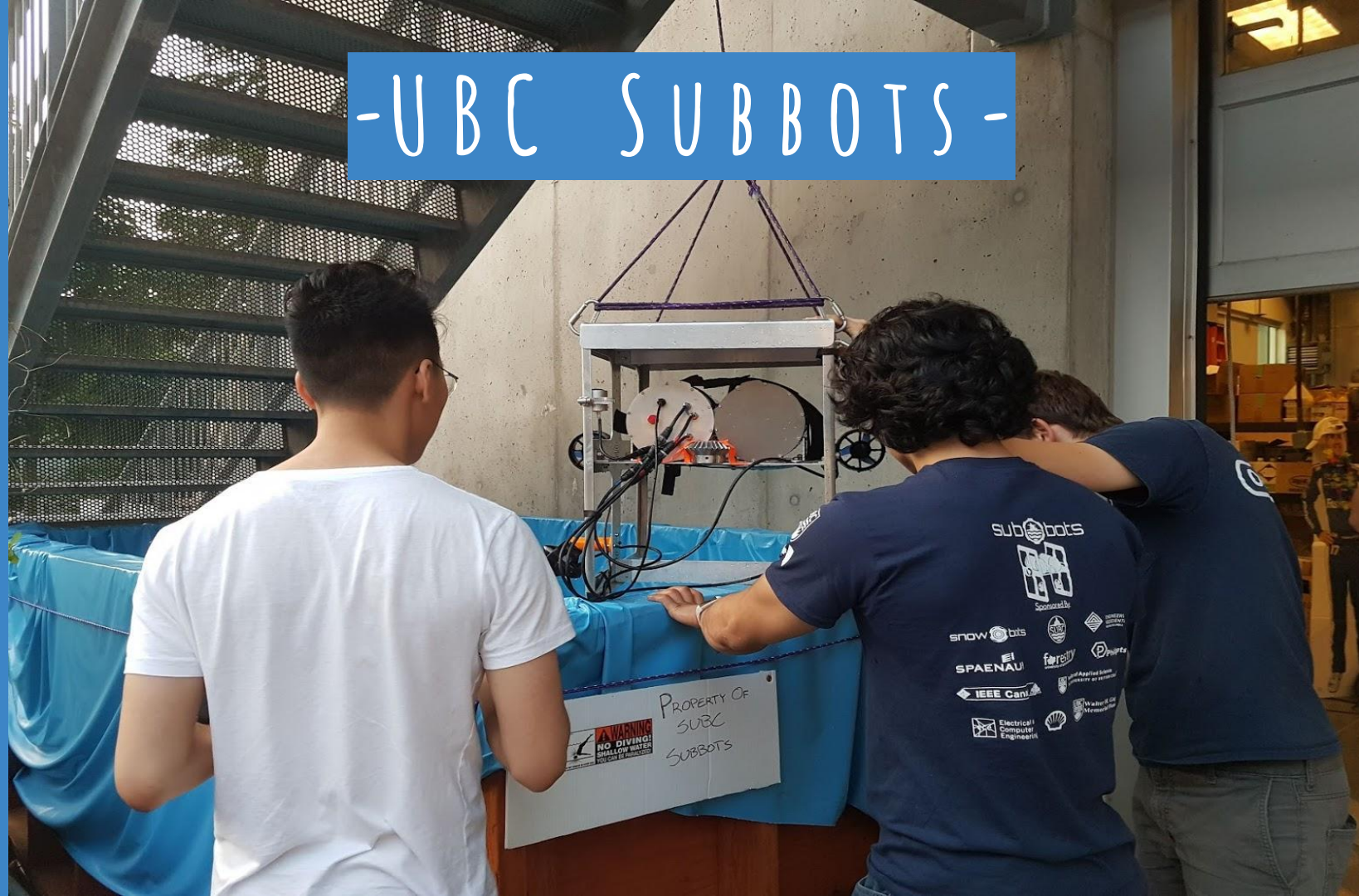


JOEL AHN

Portfolio

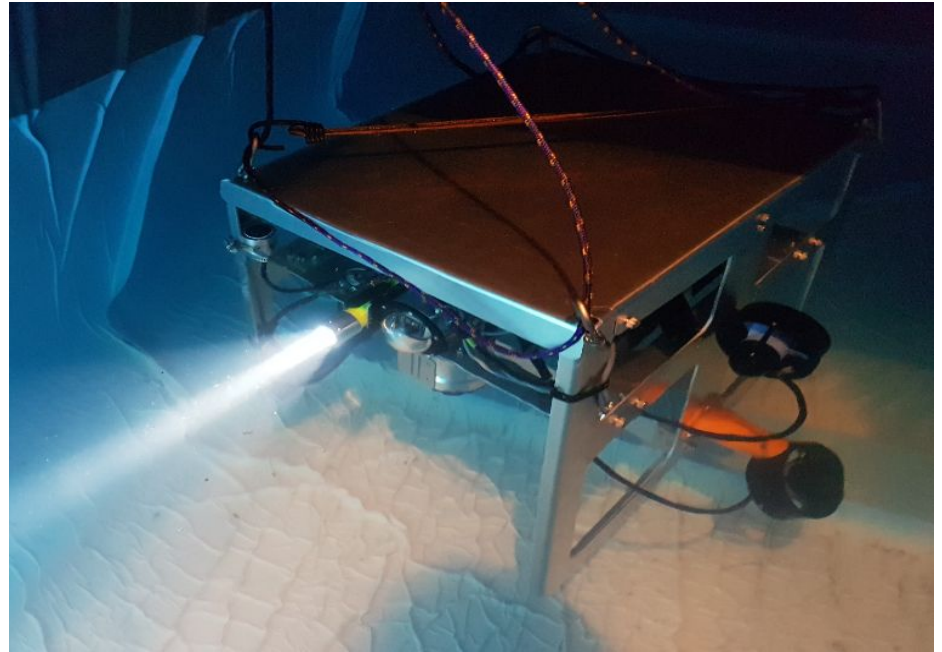
BASc. Engineering Physics - Fifth Year

# -UBC SUBBOTS-

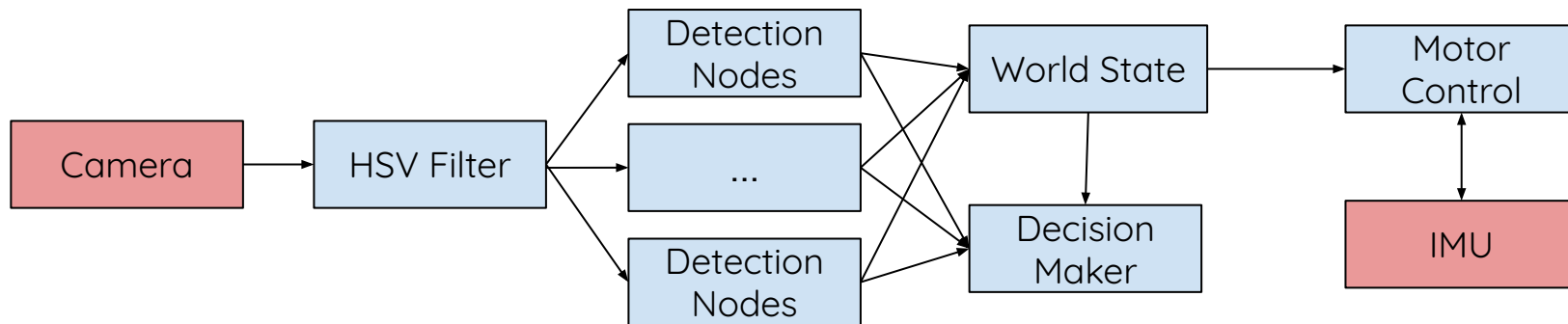


# UBC SUBBOTS

UBC SubBots dedicates its mission to designing and implementing autonomous underwater vehicles, robots capable of navigating deep waters and exploring the ingenuity of marine technology. Founded in 2017, the team competes annually at the RoboSub event, where teams bring AUVs that can accomplish everything between swimming through hoops to shooting torpedos at targets!



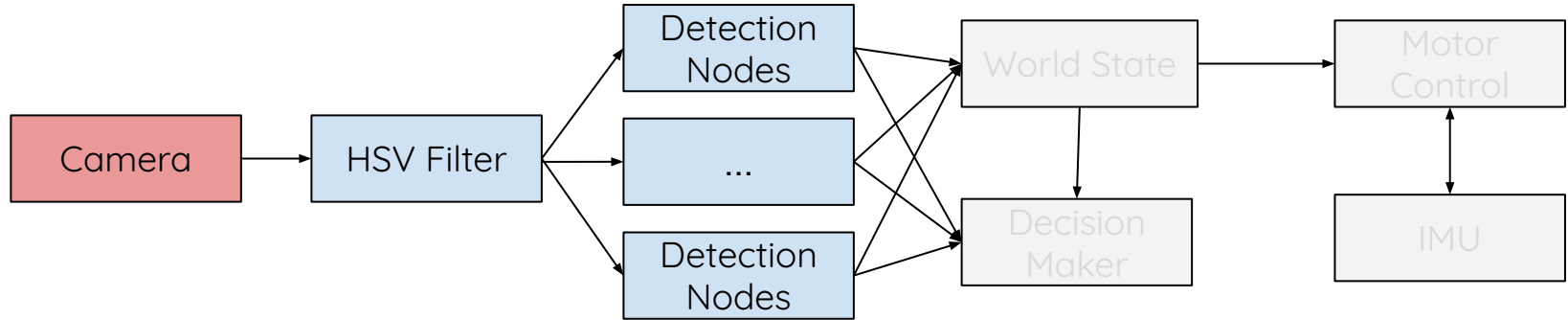
# ROS ARCHITECTURE



Part of my responsibilities on UBC SubBots was to design the pipeline architecture of our first robot Max. The Robot Operating System (ROS), a powerful platform for interconnecting processes, served as the backbone of our project. Each box represents a ROS node, each responsible for accomplishing a specialized task and communicating its due diligence.

# ROS ARCHITECTURE

## Detection Nodes

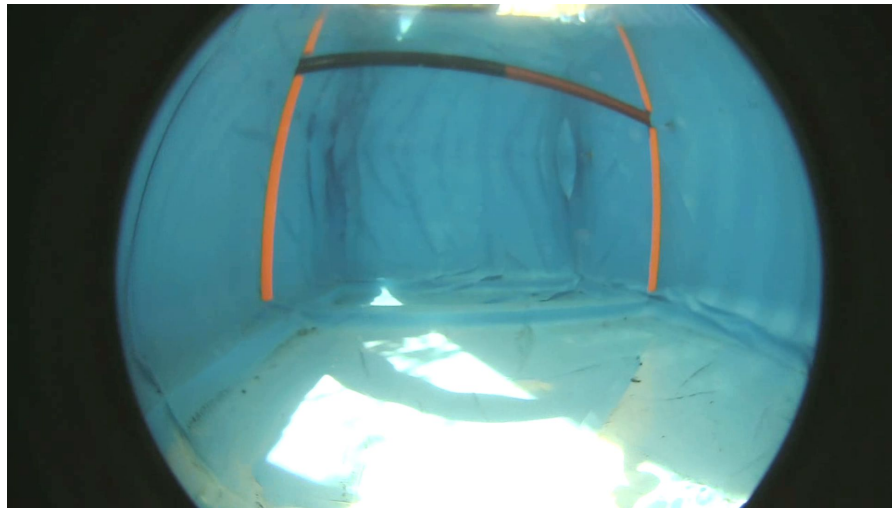


Max's primary navigation sensor was a 180° fish-eye camera. Each detection node was responsible for processing and identifying a specific object from the image that the camera node broadcasted: gate poles, die, roulette wheels. Fortunately, RoboSub published in advance the colors the objects were painted, enabling HSV filtering; and their dimensions, allowing rudimentary calculations of distance from robot to object.

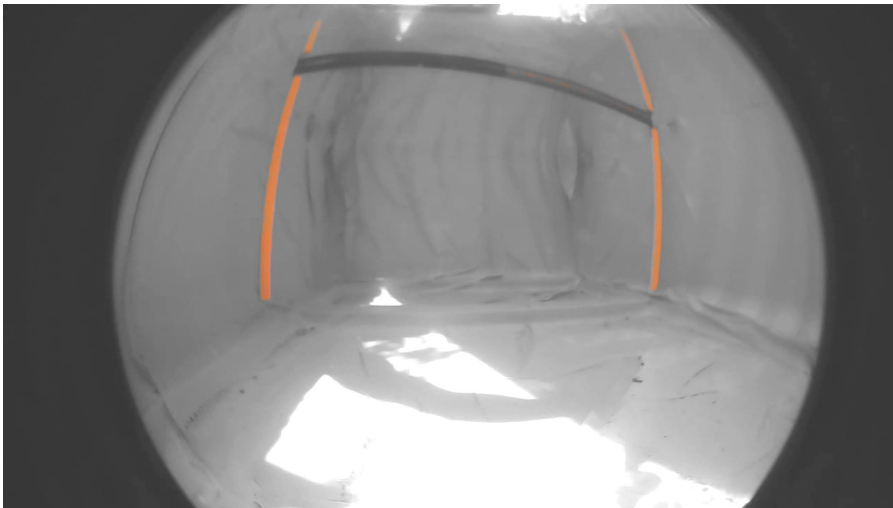
# GATE DETECTION NODE

To concretely demonstrate the detection node architecture, I helped code the gate detection Singleton class. I helped tune the HSV filtering parameters, wrote the k-clustering algorithms, and noise filtering algorithms to select the gate poles from the raw camera image.

# GATE DETECTION NODE

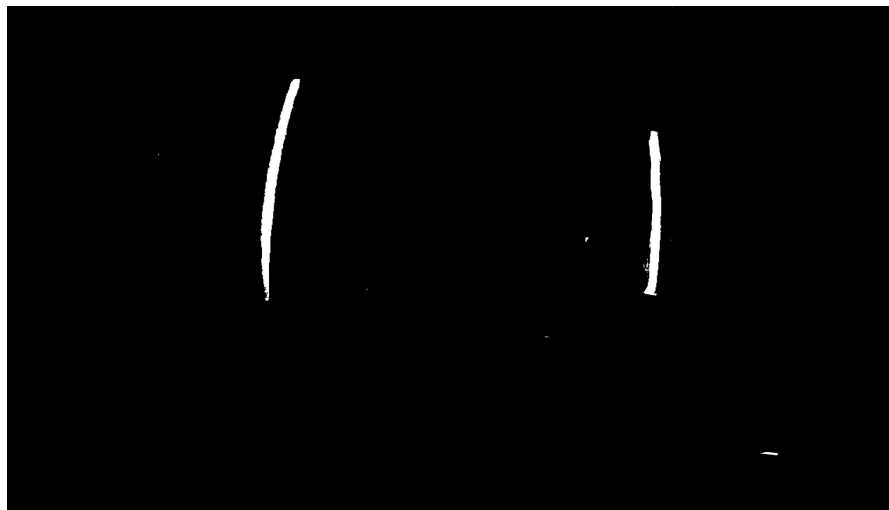


Raw Image

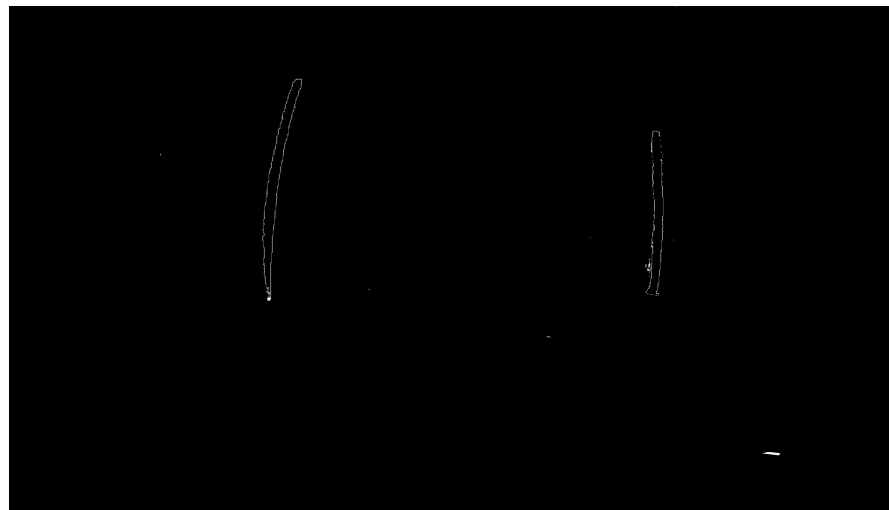


HSV-Filtered Image.  
Gate Poles detected.

# GATE DETECTION NODE



Binarized Image according to  
"similarity" of pixel HSV value to  
color filtered out.

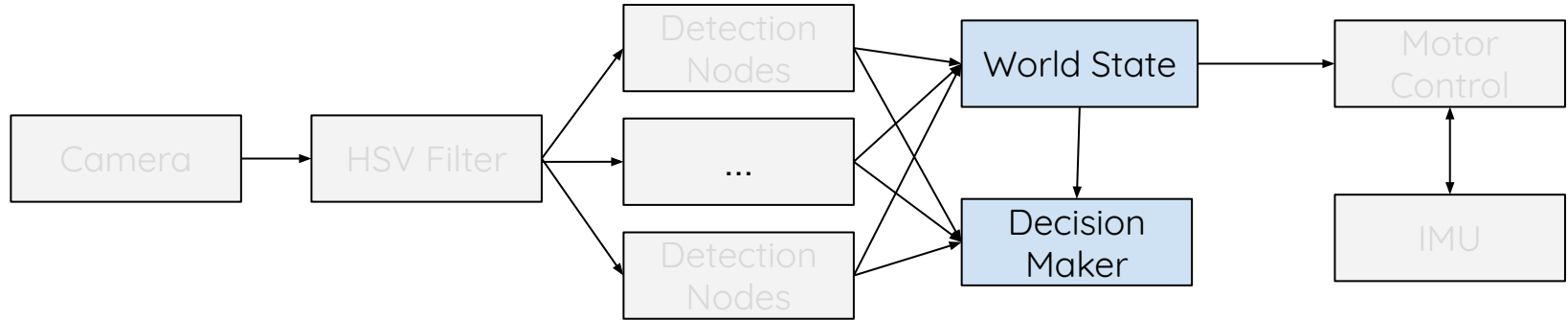


Whitespaces are k-clustered and  
non-pole-like objects are filtered  
out. Geometric algorithms (including  
fish-eye filtering) are then applied  
to judge distance from robot.



# ROS ARCHITECTURE

## World State Node

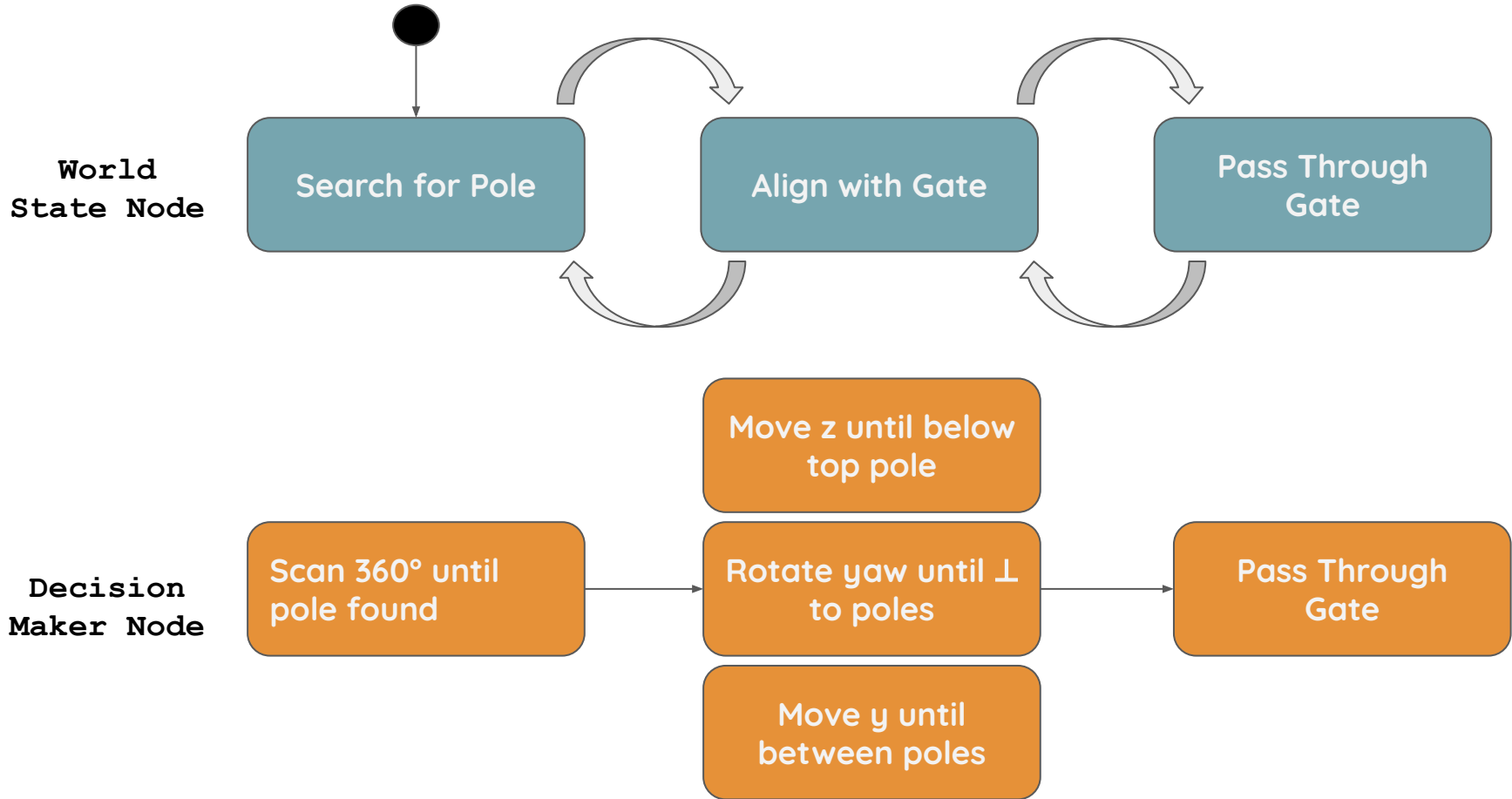


The World State Node (WSN), alongside the Decision Maker, served as the executive center of Max. Each competition portion is encapsulated as a "routine" class. WSN keeps track of the robot's sequential objectives to accomplish the routine, reporting the current one to the Decision Maker to publish navigation orders accordingly. WSN receives data from the detection nodes and readjusts objectives accordingly.

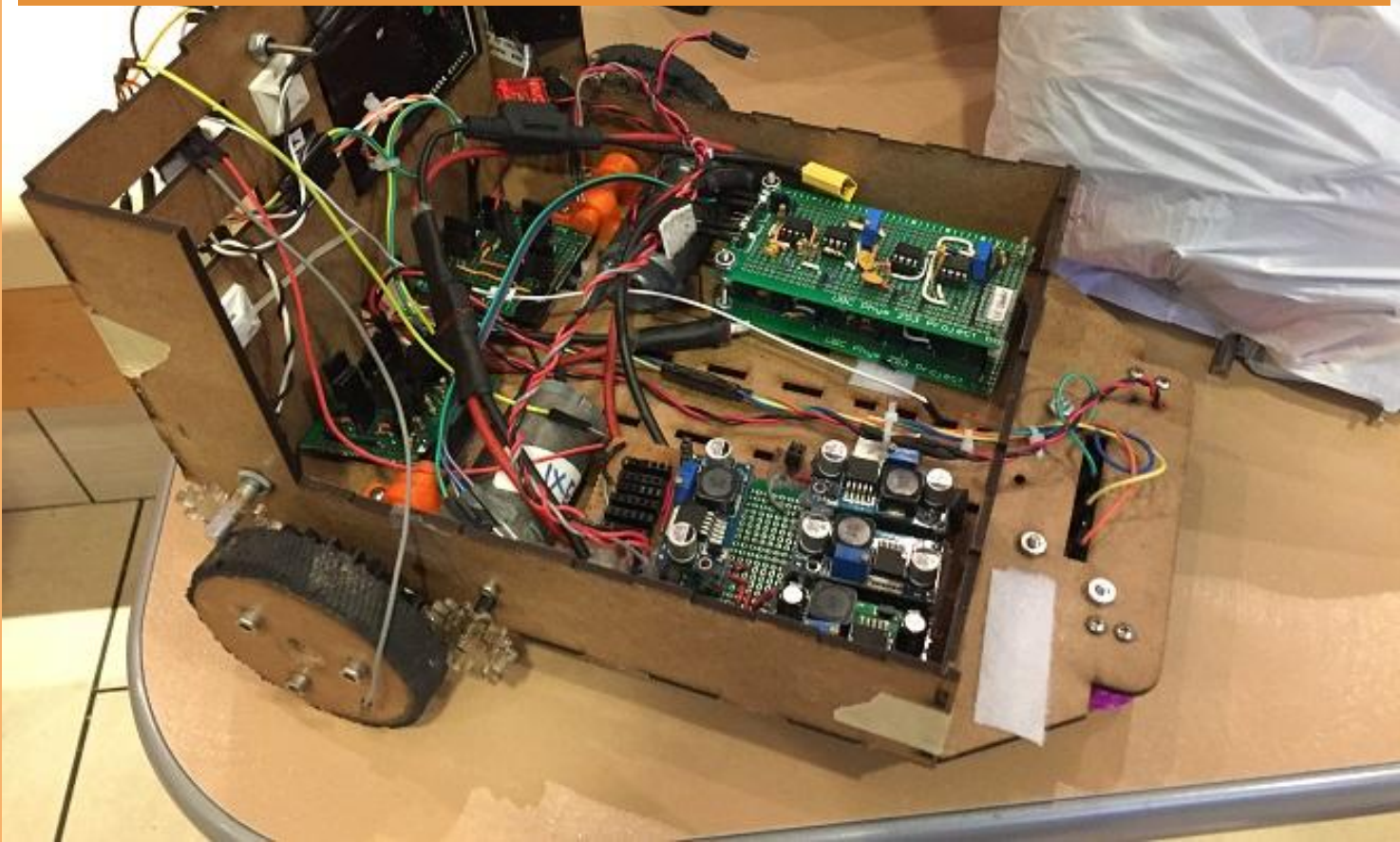
# GATE ROUTINE

The World State Node (WSN) is essentially a finite state machine where its states are Routine class objects. The Gate Routine consists of three Routine subclasses that encode a sequential objective to fulfilling the overall task of passing through the gate. Alongside writing the imaging processing algorithms to detect the gate, I wrote the world state node controller logic.

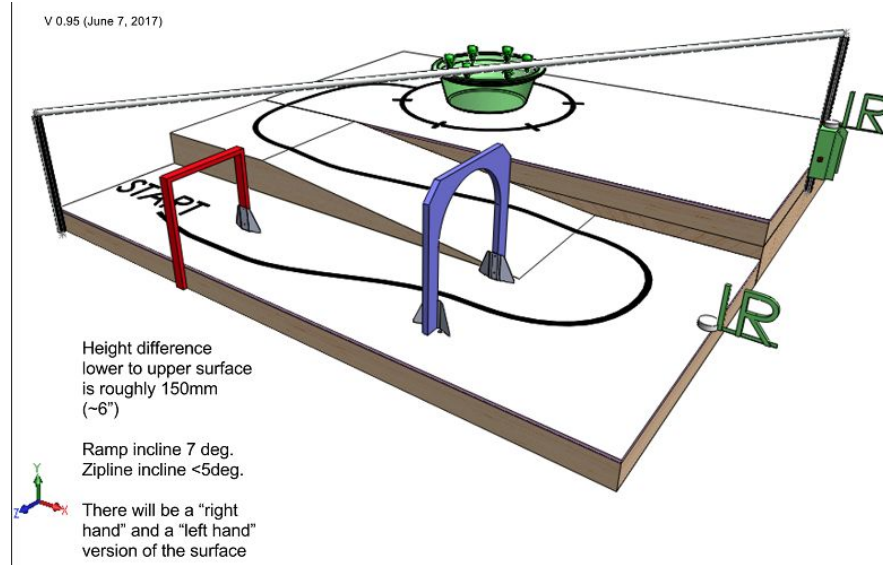
# GATE ROUTINE HI LEVEL OVERVIEW



# -ENPH 253 ROBOTICS COMPETITION: POMME DE TERROR -



# The Challenge: Mission Impossible



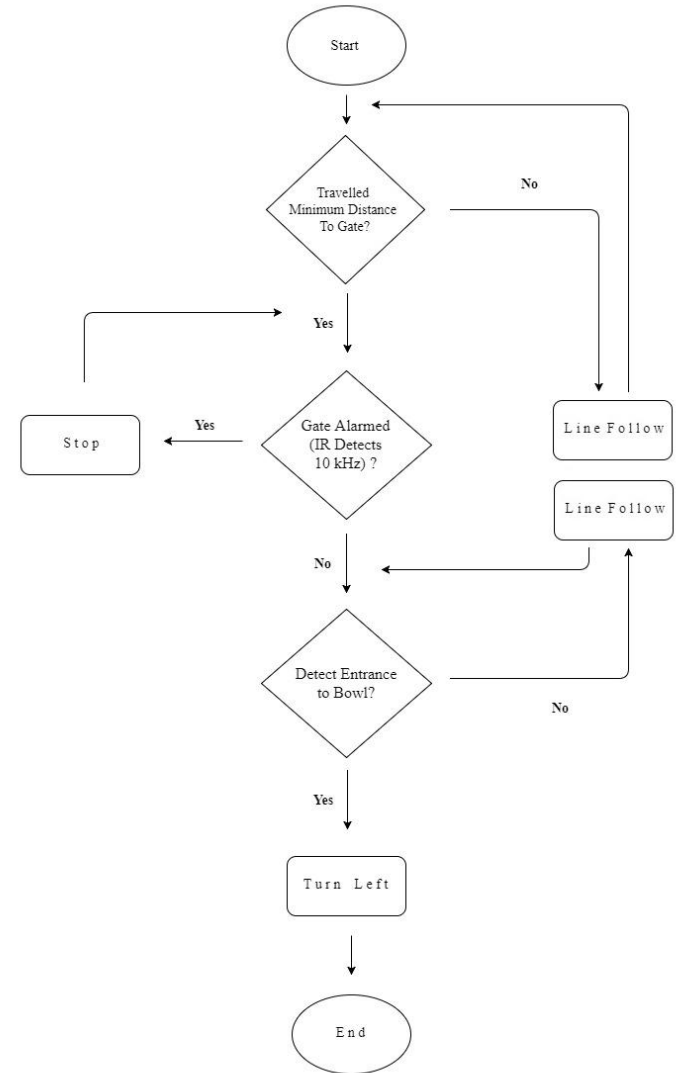
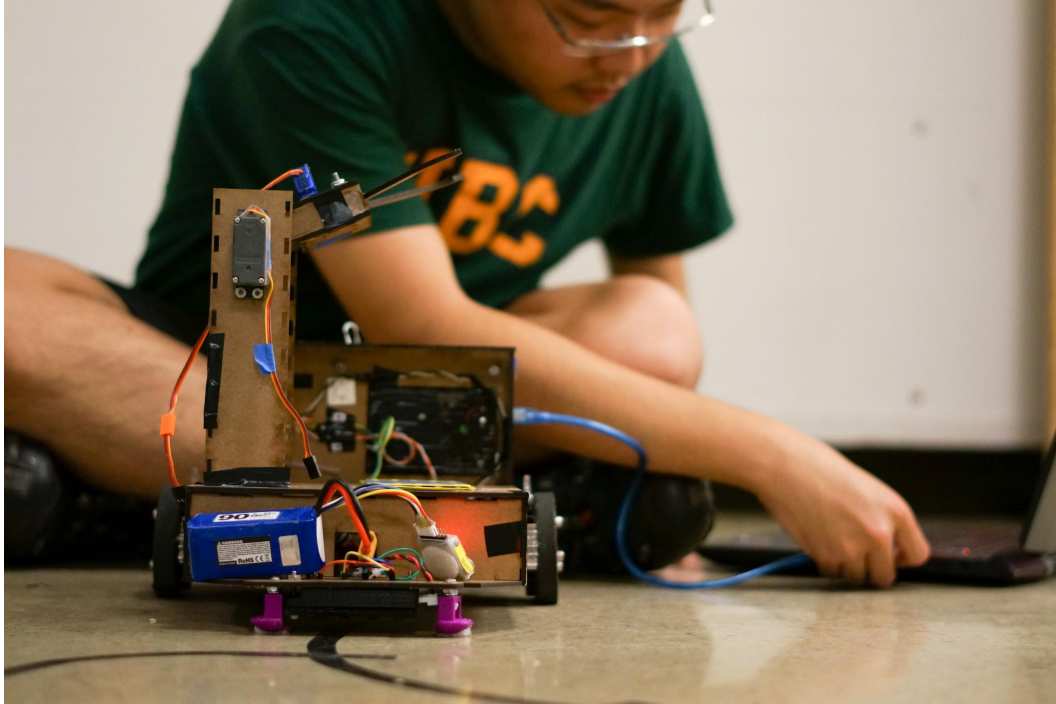
**The mission, should you choose to accept it,**

is to create an emergency rescue robot capable of navigating a black tape path;  
bypass an alarmed door when the IR signal switches from 10kHz to 1kHz; rescue the  
six operatives about to walk the plank; and deliver them down the zipline to safety.

# FSM Architecture

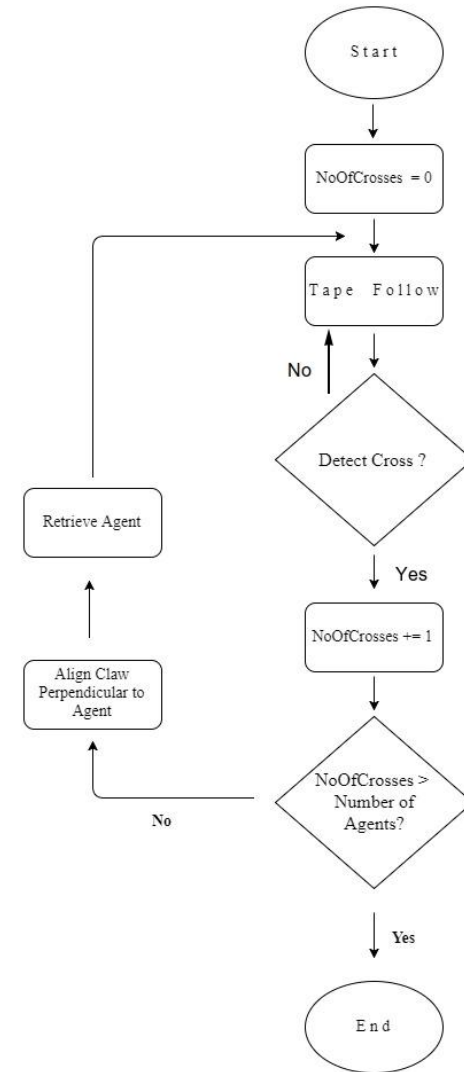
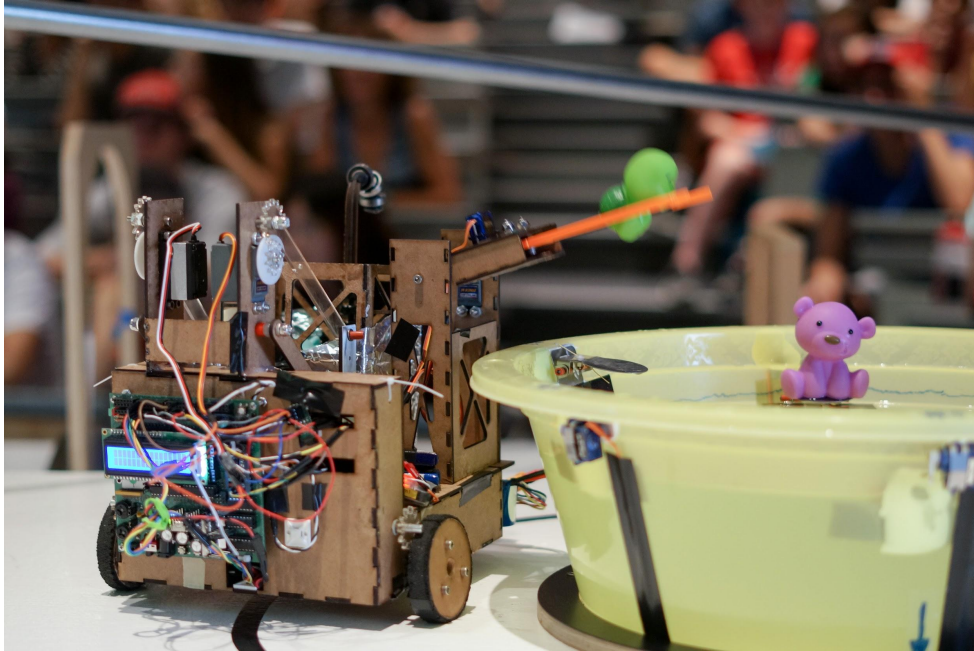
Pomme de Terror's firmware architecture utilized an overarching finite state machine, each state one portion of the challenge in a modularized class called a "routine." Each routine in turn fractalized into its own FSM. This allowed for parallel development of the control logic and enabled more stringent testing of the FSM's transition states. My tangible contributions include coding the "Pass the Gate" and "Zipline" finite state machines.

# Pass the Gate



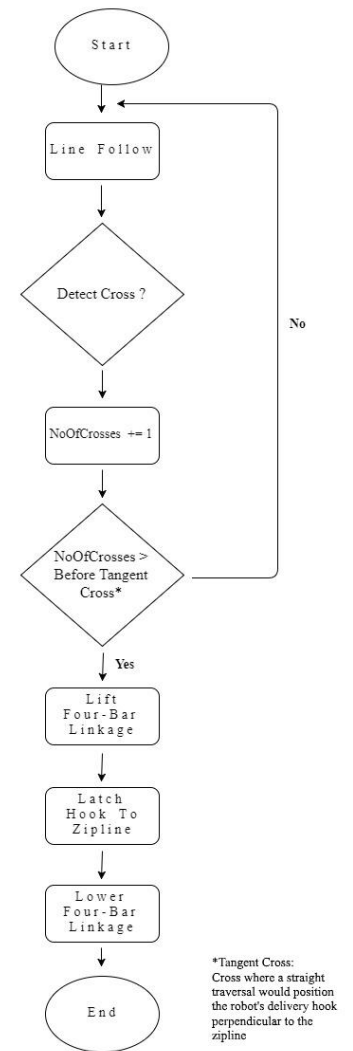
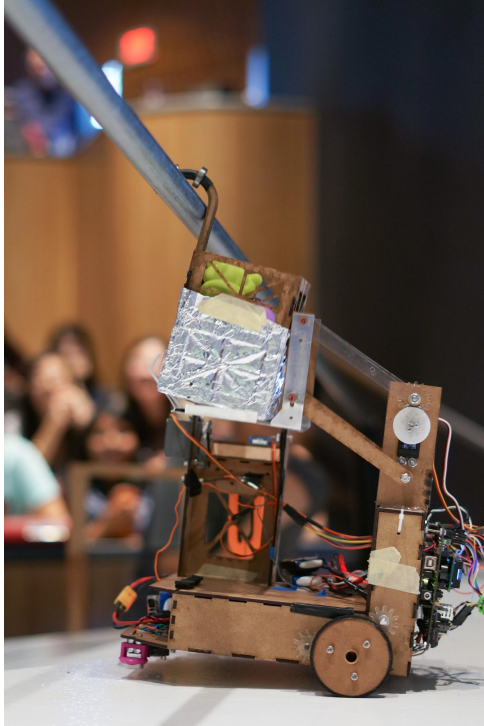


# Retrieve Agents





# Zipline

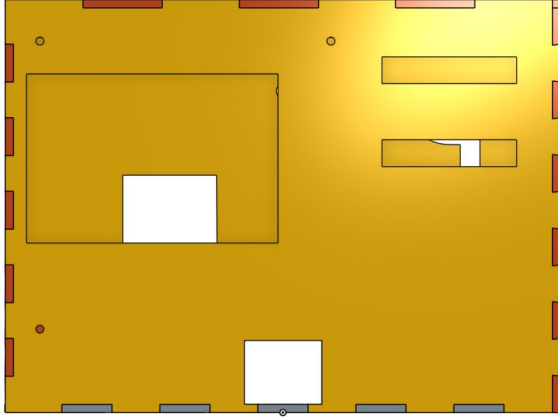


# Chassis and IR Filter

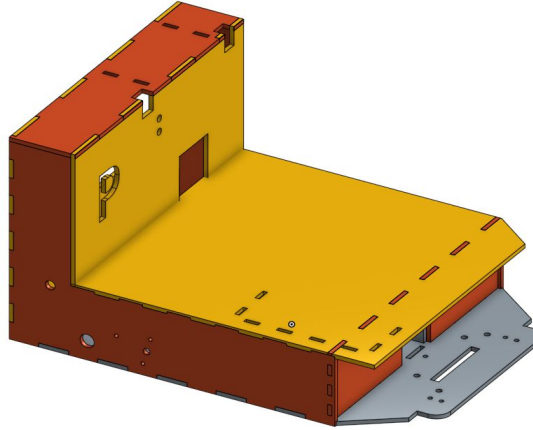
Building Pomme de Terror was stressful but enjoyable. I found the time to contribute to other components of the robot, including the chassis design and analyzing the IR filter circuit.

Through the chassis design, I learned the fundamentals of CAD design. Through the IR filter circuit, I gleaned an understanding of analog signals processing.

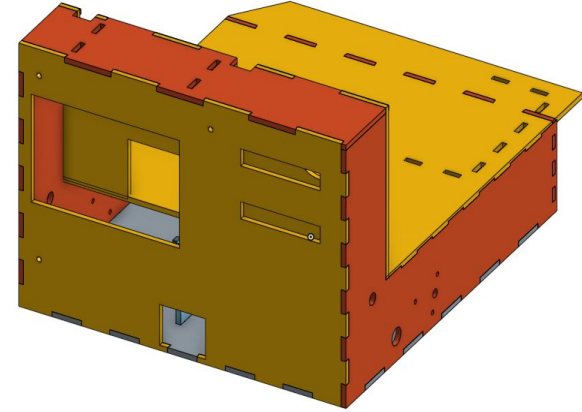
# Chassis



Large rectangular window to house MCU. Two slits to allow wires from MCU to interior PCBs.

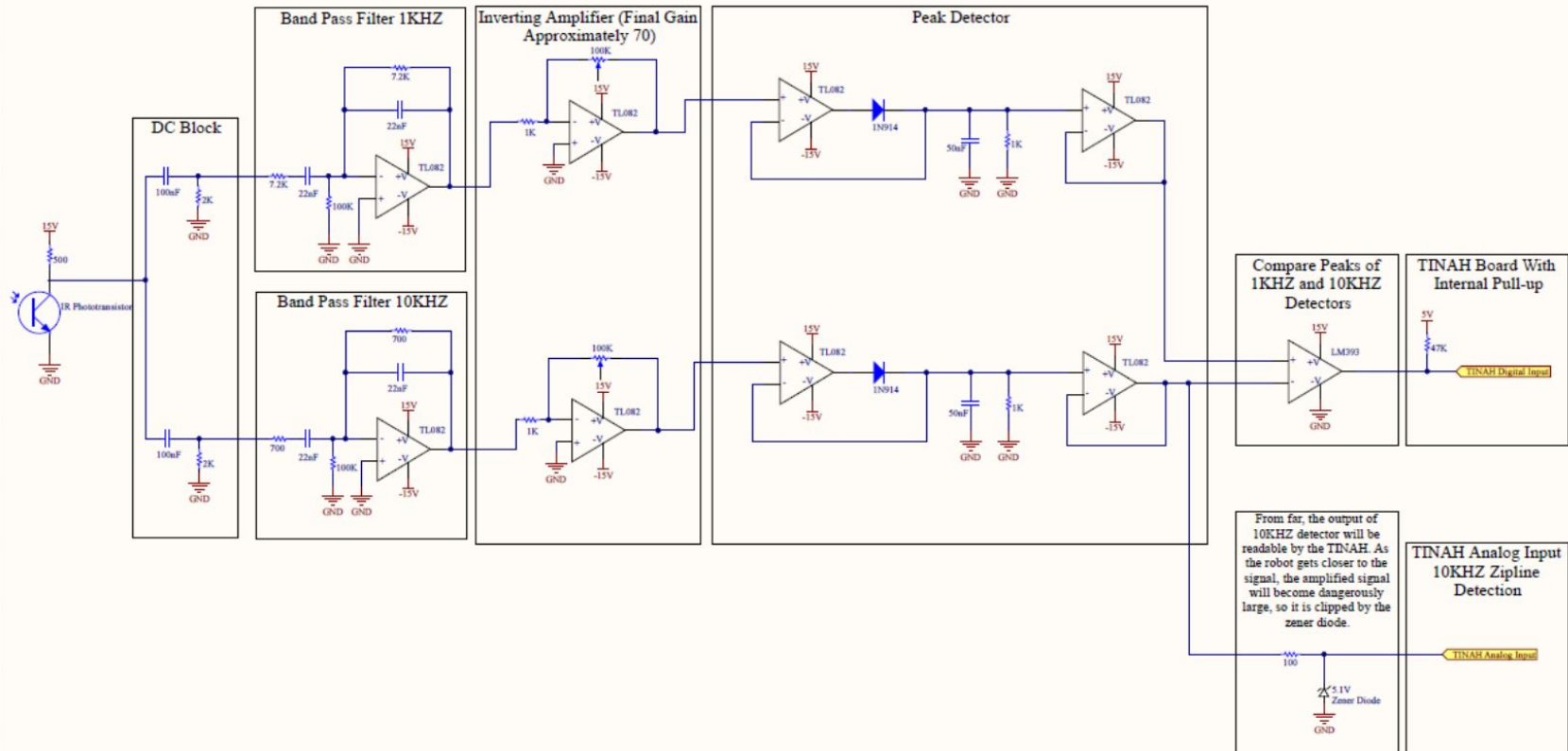


Large platform enables open and free space for modular components to be added and discarded.



Slits and holes cut for easy joint-fitting of modular components, enabling easy replacement of parts.

# IR Filter Circuit Debug



# - POMME DE TERROR -

